

Rutherford scattering 1

Find the scattering cross section for the Coulomb potential

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} = -\frac{Z\alpha\hbar c}{r}$$

Let $f(\mathbf{p})$ be the scattering amplitude for momentum transfer vector $\mathbf{p} = \mathbf{p}_i - \mathbf{p}_f$. The following formula is the Born approximation for $f(\mathbf{p})$.

$$f(\mathbf{p}) = \frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{r}}{\hbar}\right) V(\mathbf{r}) d\mathbf{r}$$

Convert to polar coordinates where $p = |\mathbf{p}|$.

$$f(\mathbf{p}) = \frac{m}{2\pi\hbar^2} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp\left(\frac{ipr \cos \theta}{\hbar}\right) V(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi$$

Substitute the Coulomb potential

$$V(r, \theta, \phi) = -\frac{Z\alpha\hbar c}{r}$$

to obtain

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{2\pi\hbar} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp\left(\frac{ipr \cos \theta}{\hbar}\right) r \sin \theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \int_0^\pi \exp\left(\frac{ipr \cos \theta}{\hbar}\right) r \sin \theta dr d\theta$$

Transform the integral over θ into an integral over u where $u = \cos \theta$ and $du = -\sin \theta d\theta$. The minus sign in du is canceled by interchanging integration limits $\cos 0 = 1$ and $\cos \pi = -1$.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \int_{-1}^1 \exp\left(\frac{ipru}{\hbar}\right) r dr du$$

Solve the integral over u .

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \left[\frac{\hbar}{ipr} \exp\left(\frac{ipru}{\hbar}\right) \right]_{u=-1}^{u=1} r dr$$

Cancel r and evaluate the limits.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \frac{\hbar}{ip} \left[\exp\left(\frac{ipr}{\hbar}\right) - \exp\left(-\frac{ipr}{\hbar}\right) \right] dr$$

Solve the integral over r .

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \frac{\hbar}{ip} \left[\frac{\hbar}{ip} \exp\left(\frac{ipr}{\hbar}\right) + \frac{\hbar}{ip} \exp\left(-\frac{ipr}{\hbar}\right) \right]_{r=0}^{r=\infty}$$

The exponentials fail to converge at the upper limit. The workaround is to go back and multiply the integrand by $\exp(-\epsilon r) \approx 1$.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \int_0^\infty \frac{\hbar}{ip} \left[\exp\left(\frac{ipr}{\hbar} - \epsilon r\right) - \exp\left(-\frac{ipr}{\hbar} - \epsilon r\right) \right] dr$$

Solve the modified integral.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \frac{\hbar}{ip} \left[\frac{1}{ip/\hbar - \epsilon} \exp\left(\frac{ipr}{\hbar} - \epsilon r\right) + \frac{1}{ip/\hbar + \epsilon} \exp\left(-\frac{ipr}{\hbar} - \epsilon r\right) \right]_{r=0}^{r=\infty}$$

Evaluate the limits. Now the exponentials vanish at the upper limit.

$$f(\mathbf{p}) = -\frac{mZ\alpha c}{\hbar} \frac{\hbar}{ip} \left(-\frac{1}{ip/\hbar - \epsilon} - \frac{1}{ip/\hbar + \epsilon} \right) \quad (1)$$

Set $\epsilon = 0$ to obtain

$$f(\mathbf{p}) = -\frac{2mZ\alpha\hbar c}{p^2}$$

Note that $\mathbf{p}$ is momentum transfer such that

$$p^2 = |\mathbf{p}|^2 = 4mE(\cos\theta - 1)$$

Hence

$$f(\theta) = -\frac{Z\alpha\hbar c}{2E(\cos\theta - 1)}$$

Calculate the cross section.

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \left[\frac{Z\alpha\hbar c}{2E(\cos\theta - 1)} \right]^2 \quad (2)$$

Note that

$$(\cos\theta - 1)^2 = 4\sin^4(\theta/2)$$

Hence equation (2) is equivalent to

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z\alpha\hbar c}{2E} \right)^2 \frac{1}{4\sin^4(\theta/2)} \quad (3)$$

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